

Ruijiang Gao

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Academic Appointment

The University of Texas at Dallas , Naveen Jindal School of Management Assistant Professor of Information Systems	2024–present
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Education

The University of Texas at Austin , McCombs School of Business Ph.D., Information, Risk and Operations Management Dissertation: “Advancing Human-AI Systems: On Robustness, Decision Making, and Beyond” Advisor: Maytal Saar-Tsechansky	2024
University of Michigan M.S., Statistics	2018
University of Science and Technology of China B.S., Statistics, School of the Gifted Young	2016

Research Interests

Human-AI decision systems; generative and agentic AI; causal policy learning; data valuation and privacy; uncertainty quantification.

Publications

An asterisk () denotes equal contribution. Journal extensions of conference papers are listed separately under papers under review.*

Refereed Journal Articles

1. Gao, Ruijiang, Mingzhang Yin, James McInerney, and Nathan Kallus. 2024. “Adjusting Regression Models for Conditional Uncertainty Calibration.” *Machine Learning*, Special Issue on Uncertainty Quantification in Machine Learning.

Refereed Conference Papers

1. Cao, Junyu*, Ruijiang Gao*, and Esmaeil Keyvanshokoo*. 2025. “HR-Bandit: Human-AI Collaborated Linear Recourse Bandit.” In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*.
2. Gao, Ruijiang, and Mingzhang Yin. 2025. “Confounding-Robust Deferral Policy Learning.” In *Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI)*.
3. Gao, Ruijiang, Mingzhang Yin, and Maytal Saar-Tsechansky. 2024. “SEL-BALD: Deep Bayesian Active Learning for Selective Labeling with Instance Rejection.” In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.
4. Gao, Ruijiang, and Himabindu Lakkaraju. 2023. “On the Impact of Algorithmic Recourse on Social Segregation.” In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
5. Wang, Zhendong*, Ruijiang Gao*, Mingzhang Yin*, Mingyuan Zhou, and David M. Blei. 2023. “Probabilistic Conformal Prediction Using Conditional Random Samples.” In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS)*.
6. Gao, Ruijiang, Max Biggs, Wei Sun, and Ligong Han. 2022. “Enhancing Counterfactual Classification Performance via Self-Training.” In *Proceedings of the 36th AAAI Conference on Artificial Intelligence (AAAI)*.

7. Han, Ligong, Martin Renqiang Min, Anastasis Stathopoulos, Yu Tian, Ruijiang Gao, Asim Kadav, and Dimitris N. Metaxas. 2021. "Dual Projection Generative Adversarial Networks for Conditional Image Generation." In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 14438–14447.
8. Gao, Ruijiang, Maytal Saar-Tsechansky, Maria De-Arteaga, Ligong Han, Min Kyung Lee, and Matthew Lease. 2021. "Human-AI Collaboration with Bandit Feedback." In *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*.
9. Gao, Ruijiang, and Maytal Saar-Tsechansky. 2020. "Cost-Accuracy Aware Adaptive Labeling for Active Learning." In *Proceedings of the 34th AAAI Conference on Artificial Intelligence (AAAI)*, 34(3): 2569–2576.
10. Han, Ligong, Ruijiang Gao, Mun Kim, Xin Tao, Bo Liu, and Dimitris N. Metaxas. 2020. "Robust Conditional GAN from Uncertainty-Aware Pairwise Comparisons." In *Proceedings of the 34th AAAI Conference on Artificial Intelligence (AAAI)*, 10909–10916.
11. Han, Ligong, Yang Zou, Ruijiang Gao, Lezi Wang, and Dimitris N. Metaxas. 2019. "Unsupervised Domain Adaptation via Calibrating Uncertainties." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 99–102.

Papers Under Review and in Revision

1. Gao, Ruijiang, and Steven Chong Xiao. "When AI Analysts Disagree: Agentic Multiverse Analysis and the Robustness of Empirical Research." **Job Market Paper**. Under review at *Information Systems Research*. Earlier version circulated as "Nonstandard Errors in AI Agents."
2. Gao, Ruijiang, Maria De-Arteaga, and Maytal Saar-Tsechansky. "Learning Complementary Policies for Human-AI Deferral Collaboration." *Management Science*, second-round review; revision submitted July 2026.
3. Yang, Yunxuan (Michael), Ruijiang Gao, and Zhiqiang (Eric) Zheng. "Sell Data to AI Algorithms Without Revealing It: Secure Data Valuation and Sharing via Homomorphic Encryption." *Management Science*, invited major revision, August 2026.
4. Cao, Junyu*, Ruijiang Gao*, Esmaeil Keyvanshokoh*, and Jianhao Ma*. "LIBRA: Language Model Informed Bandit Recourse Algorithm for Personalized Treatment Planning." *Manufacturing & Service Operations Management*, under first-round review.
5. Biggs, Max*, Ruijiang Gao*, and Wei Sun*. "Loss Functions for Discrete Contextual Pricing with Observational Data." *Manufacturing & Service Operations Management*, under review.
6. Yin, Mingzhang, Ruijiang Gao, and Ziwei Cong. "Generative Personalization with Textual and Non-textual Data" (formerly "Personalizing Language Models for Generative Targeting"). *Marketing Science*, under review following a reject-and-resubmit decision.
7. Bian, Zeyu*, Max Biggs*, Ruijiang Gao*, and Zhengling Qi*. "Beyond Demand Estimation: Consumer Surplus Evaluation via Cumulative Propensity Weights." *Operations Research*, under review.
8. Deng, Junwei, Xirui Jiang, Shiyuan Zhang, Shichang Zhang, Himabindu Lakkaraju, Ruijiang Gao, Chris Donahue, and Jiaqi W. Ma. "Computational Copyright: Towards a Royalty Model for Music Generative AI." *Manufacturing & Service Operations Management*, under review.

Working Papers

1. Chen, Zenan*, Ruijiang Gao*, and Yingzhi Liang*. "Revealing AI Reasoning Increases Trust but Crowds Out Unique Human Knowledge." Working paper. Presented at WISE 2025.
2. Gao, Ruijiang, and Hao Pang. "Ninety-Eight Percent Talk: Digital Twins of the FOMC and Monetary Policy Surprises." Working paper. Presented at SCECR 2026.
3. Lee, Myung Jin, Ruijiang Gao, and Esmaeil Keyvanshokoh. "Sequential Feature Acquisition for Treatment Assignment under Partial Observability." Working paper. Accepted for presentation at the INFORMS Workshop on Data Science 2026.

4. Yang, Yunxuan (Michael), Ruijiang Gao, and Zhiqiang (Eric) Zheng. "Insight without Sight: LLM Inference with Private Data." Working paper. Accepted for presentation at the INFORMS Workshop on Data Science 2026.
5. Qing, Zhining, Ruijiang Gao, and Mingzhang Yin. "Evaluating Agents When Experts Know More: A Doubly-Valid and Doubly-Sharp Approach." Working paper. Accepted for presentation at the INFORMS Workshop on Data Science 2026.

Selected Honors and Funding

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| • Best Student Paper Award, Workshop on Information Technologies and Systems (WITS) | 2025 |
| • Best Paper Award, INFORMS Workshop on Data Science | 2025 |
| • Nunamaker-Chen Dissertation Award, Runner-Up, INFORMS Information Systems Society | 2025 |
| • New Faculty Research Grant, UT Dallas, Principal Investigator, \$25,000 | 2025 |
| • Best Paper Runner-Up Award, WITS | 2024 |
| • Best Student Paper Award, Conference on Information Systems and Technology (CIST) | 2022 |
| • Ph.D. Incubator Special Recognition Award Finalist, INFORMS Advances in Decision Analysis | 2022 |
| • Graduate School Continuing Fellowship, UT Austin | 2022 |
| • Professional Development Award and Good Systems Student Conference Grant, UT Austin | 2020 |
| • Graduate School Provost Fellowship, UT Austin | 2018 |

Teaching

The University of Texas at Dallas 2025–2026

- ITSS 3300, Information Technology for Business: two sections in Spring 2025 (127 students), three in Fall 2025 (193 students), and three in Fall 2026 (in progress).
- Across the five completed sections, the unweighted mean of section-level median ratings was 4.49/5 for overall instructor effectiveness and 4.45/5 for overall course quality.

The University of Texas at Austin

- Instructor, MIS 301, Introduction to Information Technology Management, Spring 2022 (60 students).
- Teaching assistant: Strategy for Networked Economy; Predictive Analysis and Data Mining; Database Management; Introduction to Data Science.

Selected Presentations

- "Nonstandard Errors in AI Agents": Taiwan Summer Workshop on Information Management, 2026.
- "Ninety-Eight Percent Talk: Digital Twins of the FOMC and Monetary Policy Surprises": SCECR, 2026.
- "Revealing AI Reasoning Increases Trust but Crowds Out Unique Human Knowledge": WISE, 2025.
- "Human-AI Collaborative Decision Making": UT Dallas Statistics Seminar, 2024.
- "SEL-BALD": WITS, 2024.
- "Learning Complementary Policies for Human-AI Teams": CIST and INFORMS Workshop on Data Science, 2022; SCECR, 2021.
- "Loss Functions for Discrete Contextual Pricing with Observational Data": INFORMS Revenue Management and Pricing Conference (spotlight) and Advances in Decision Analysis Conference, 2022; INFORMS Annual Meeting, 2021.

Professional Service

Editorial board: Decision Sciences, Agentic AI and Human-Agent Collaboration in Business department, 2026–present.

Journal reviewing: Management Science; Information Systems Research; MIS Quarterly; Operations Research; INFORMS Journal on Computing; IEEE Transactions on Pattern Analysis and Machine Intelligence; Scientific Reports.

Conference program committees and reviewing: NeurIPS, ICML, ICLR, AISTATS, AAAI, FAccT, WACV, WITS, ICIS, and INFORMS Workshop on Data Science.

Research and Industry Experience

- Netflix Research, Machine Learning Research Intern Summer 2023
Hosts: James McNerney and Nathan Kallus.
- Harvard Business School, Visiting Researcher Summer 2022
Host: Himabindu Lakkaraju.
- IBM Thomas J. Watson Research Center, Research Intern Summers 2020 and 2021
Hosts: Wei Sun, Max Biggs, and Markus Etzl.
- Tencent, Data Scientist Intern 2018
- Amazon, Business Intelligence Engineer Intern 2017

Patent Applications

1. Gao, Ruijiang, Wei Sun, Max Biggs, Markus Etzl, and Youssef Drissi. “Counterfactual Self-Training.” U.S. Patent Application 17/402,367, 2023.
2. Gao, Ruijiang, Wei Sun, Max Biggs, Youssef Drissi, and Markus Etzl. “Imputing Counterfactual Data to Facilitate Machine Learning Model Training.” U.S. Patent Application 17/654,617, 2023.